Near-Infrared Spectroscopy for On-Line Monitoring of Lube Base Oil Processes

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Near-infrared (NIR) spectroscopy has been utilized for monitoring the pour point of the lube base oil (LBO) process. Using partial least-squares (PLS) regression, necessary spectral features were successfully extracted and correlated to the reference pour-point data. In a preliminary laboratory feasibility study, it was found that PLS calibration performance was largely governed by the accuracy of the reference pour-point analysis. The same NIR spectroscopic methodology was moved to monitor the pour point continuously in an on-line manner. At this time, the NIR spectrometer was calibrated with the existing on-line pour-point analyzer, which was more accurate. In comparison with the on-line pour-point analyzer, NIR spectroscopy provided faster and more repeatable analytical data. With the help of NIR analysis, LBO process variations during product switch can be detected earlier, and necessary process-control strategy can be applied to improve the process efficiency.

Index Headings: **Near infrared; NIR; Partial least squares; PLS; Pour point; Lube base oil; On-line process monitoring.**

INTRODUCTION

Lube base oil (LBO)[1–3] is one of the most popular petroleum products. LBO is a basic feed stock for many different types of lubricants, which are produced by blending LBO with the necessary amounts of different functional additives such as anti-oxidants and viscosity enhancers. LBO is mainly composed of iso-paraffins (saturated hydrocarbons with a branched structure) with different chain lengths (molecular weights). Generally, when the chain length of LBO becomes longer, then its viscosity becomes higher. There are several physical properties determining the quality of the final LBO product such as viscosity, viscosity index, and pour point. Among them, the pour point is an important property that determines the lube quality in cold weather conditions. The pour point corresponds to the temperature at which a sample becomes stagnant (no flow). It is related to the number of carbon atoms (molecular weight), as well as to the paraffin structure of LBO. When the molecular weight is lower and the structure is more branched (more iso-paraffin present), the pour point will be lower. Contrarily, when the molecular weight is higher and the structure is more linear (more normal paraffin, so-called wax, present), the pour point will be higher (no flow at even higher temperatures). Conventionally, the pour point is measured by the method of ASTM D97.[4] After preliminary heating, the sample is cooled at a specified rate and examined at intervals of 3 °C for flow characteristics. When the sample shows no movement in the test jar (horizontal position) for five seconds, then the corresponding temperature is recorded as the pour point. However, it requires long analysis time (30~50 min) due to the repetition of checking the flow and decreasing the temperature; also, the accuracy is limited by the temperature decrease interval of 3 °C.

Near-infrared (NIR) spectroscopy[5] has been studied and implemented for the on-line measurement of the pour point of LBO. Four different grades of LBO products were evaluated in this study. The collected NIR spectra contained the necessary analytical information to identify the pour point, such as spectral variations related to different chain lengths of normal linear hydrocarbons. Partial least-squares (PLS) regression[6,7] was used to extract relevant spectral information and build a calibration model. Two different methods of reference pour-point measurements have been accomplished. First, the pour point was measured with a 3 °C decrease interval from 0 °C as described in the standard ASTM method. Second, the pour point was measured three times from 0, −1, and −2 °C with the same 3 °C decrease interval. Out of three individual pour-point measurements from different initial temperatures, the highest temperature was determined as the final pour point. The PLS calibration results with the second reference method (the more precise measurement) showed better correlation in comparison to that using the original ASTM standard method. The more accurate laboratory measurement resulted in the better PLS calibration results. The more accurate pour point can be measured by examining the narrower temperature interval using the second method.

After the preliminary study, an on-line NIR analyzer with the necessary sampling system was implemented. The on-line NIR analyzer was calibrated with the use of the conventional on-line pour-point analyzer and prediction performance was continuously compared. The on-line NIR prediction data showed excellent correlation with those from the conventional analyzer and also showed precise measurement.

EXPERIMENTAL

Instrument and Apparatus. ***Near-Infrared Spectra .*** Near-infrared spectra were collected over the 1100 to 2500 nm spectral region with a Foss NIRSystems model 6500 spectrometer (Foss NIRSystems, Silver Spring, MD) equipped with a tungsten halogen lamp, PbS detector, and fiber-optic transflectance probe. The resolution of the collected spectra was 10 nm with 2 nm data intervals. The fiber-optic probe consisted of concentric rings of illuminating fibers (inner core, 210 fibers), receiving fibers (outer ring, 210 fibers), and a reflecting mirror. The size of the fiber optic probe was 2.54 cm (outer diameter) and

Received 29 August 2002; accepted 23 December 2002.
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0003-7028/03/5705-0545$2.00/0
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15.2 cm (length). The length of the optical fibers from the probe to the spectrometer was 2 m. The distance between the optical fibers and the reflecting mirror was 0.5 cm, providing an actual pathlength of 1 cm. NIR spectra were collected by positioning the fiber-optic probe in each sample, which was held in a sealed bottle. Each NIR sample spectrum consisted of 16 coadded scans (acquisition time: 10 s).

Procedures. ***Sample Preparation.*** Forty-three lube base oil (LBO) samples for the calibration data set (for the laboratory feasibility study) were obtained over 45 days at SK Corporation (Ulsan, Korea).[8] Samples were collected over this period to give more process-related variations into the data set, which would result in more robust calibration models. Immediately after collection, samples were sealed and stored in a refrigerator at 4 °C to prevent evaporation of the hydrocarbons. The pour point of each sample was measured by the ASTM D97 method, as well as by the same method with three different initial temperatures, as described in the Introduction. Pure ***n***-hexane, ***n***-octane, ***n***-decane, ***n***-dodecane, and ***n***-hexadecane were purchased from common suppliers.

Partial Least-Squares Regression. All data processing, including PLS regression and second-derivative algorithms, was accomplished by using NSAS ver. 3.3 software (Foss NIRSystems, Silver Spring, MD).

RESULTS AND DISCUSSION

Preliminary Examination. The pour point of lube base oil is mostly governed by the quantity as well as the molecular weight of normal linear hydrocarbon (so-called wax). As the chain length and its content of normal linear hydrocarbons increase, the pour point increases (easy to stagnate at higher temperature). Therefore, to determine the pour point by NIR spectroscopy, qualitative spectral differences between different chain lengths of normal linear hydrocarbons in the NIR region are required, since these spectral features are expected to be very similar due to their similar molecular structure. If there is no significant spectral variation among normal linear hydrocarbons, the pour point cannot be predicted with NIR spectroscopy. To examine the spectral features, NIR spectra of five different chain lengths of normal linear hydrocarbons (***n***-hexane (C_6), ***n***-octane (C_8), ***n***-decane (C_{10}), ***n***-dodecane (C_{12}), and ***n***-hexadecane (C_{16})) were collected. The resulting spectra are shown in Fig. 1. For clear examination, the 1100–1300 nm range was also enlarged (bottom plot). All the spectra were offset for clarity. The overall shapes of all the spectra are similar to each other; however, reasonable spectral variations are observed in the 1100–1600 and 1850–2150 nm range, while the most systematic spectral variations are observed from 1120 to 1270 nm. As increasing the chain length in normal linear hydrocarbon corresponds to increasing the number of CH_2 groups with a fixed number of CH_3 groups, the CH_2 second overtone band at 1210 nm becomes more dominant while the CH_3 second overtone band at 1194 nm becomes less dominant. In the ***n***-hexane spectrum, both the 1194 and the 1210 nm bands are observed; however, the 1194 nm band is clearly decreased in the spectrum of ***n***-hexadecane. Overall, fairly clear and systematic spectral variations are observed among different normal linear hydrocarbons, as shown in Fig. 1.

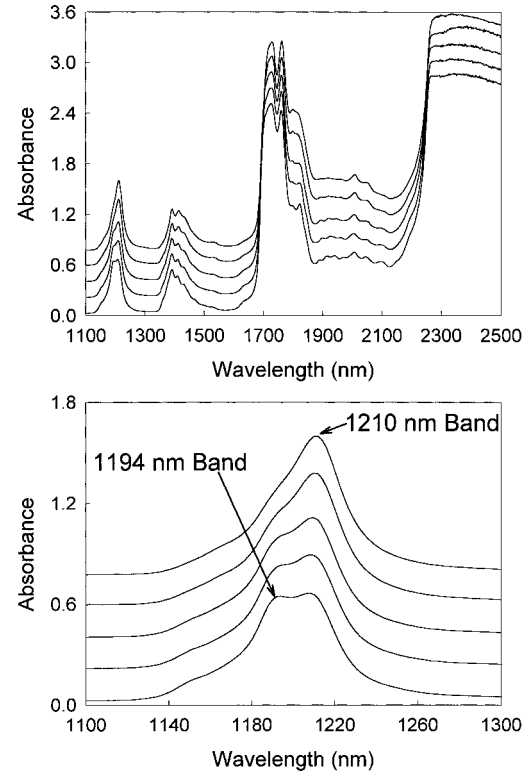

FIG. 1. NIR spectra of normal linear hydrocarbons. The top plot shows the overall NIR range (1100–2500 nm) and the bottom plot shows the 1100–1300 nm range for magnification. In each plot, spectra from bottom to top correspond to *n*-hexane, *n*-octane, *n*-decane, *n*-dodecane, and *n*-hexadecane (offset for clarity).

Figure 2 (top) shows all the NIR spectra of the data set (including different grades of product) that were used to build a calibration model. The most useful spectral information is located in the 1100–1580 and 1870–2140 nm spectral ranges. The 1580–1870 and 2140–2500 nm ranges contain no useful spectral information due to the strong absorption of the NIR radiation from the long optical pathlength; therefore, these two ranges are excluded from further analysis in this study.

Raw NIR spectra of the data set show slight baseline variations, so it is hard to examine the product-related spectral variations. Therefore, all original raw spectra were converted to second-derivative spectra to enhance the spectral features. It is well known that second derivatives of raw spectra help to enhance the spectral features and remove (or at least reduce) baseline variation.[9] Figure 2 (middle) shows the second derivative spectra of the data set. The baseline variations are clearly removed, while keeping the same qualitative information. However, as shown in these second-derivative spectra, only small spectral variations are observed in the 1100–1580 nm region. To examine the spectral variations more clearly, each second-derivative spectrum was subtracted from the average spectrum of the data set. By using these difference spectra, the spectral variations can be clearly enhanced. The results are shown in Fig. 2 (bottom). The spectral variations are clearly observed in the 1100–1580 and 1870–2140 nm region. The 1100–1300 nm region,

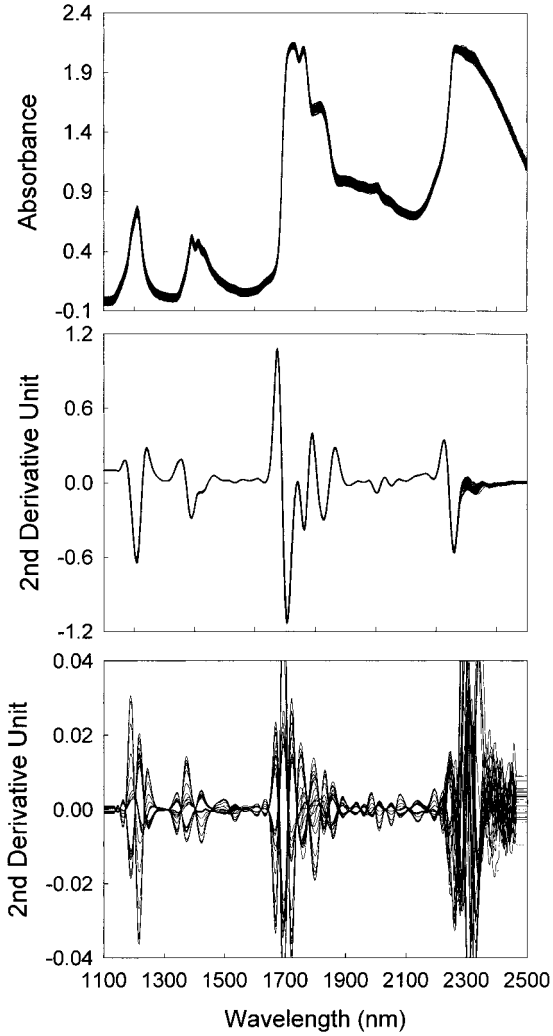

FIG. 2. Raw NIR spectra of the data set (top), corresponding second-derivative spectra (middle), and second-derivative difference spectra subtracted from the average spectrum of the data set (bottom).

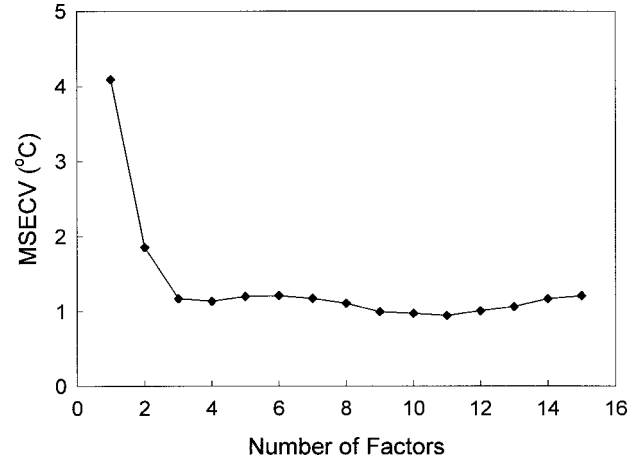

FIG. 3. MSECV plotted as a function of the number of PLS factors used to determine the pour point within the 1100–1580 nm range (Experiment B).

especially, shows the greatest spectral variations among the different LBO products. This result complies with that of pure normal linear hydrocarbon, in that the most selective spectral information for different normal paraffin structures is observed in the 1100–1300 nm region. The 1100–1300 nm region provides the most selective spectral information for the measurement of the pour point of LBO.

Partial Least-Squares Calibration (Laboratory Feasibility Study). As shown in Fig. 2, the overall absorption bands show reasonable spectral variations with different grades of samples; however, the resulting NIR spectra are highly overlapped due to the compositional complexity of the LBO samples. Therefore, multivariate calibration methods[6,7] such as PLS regression are required to resolve these mixtures, as has been applied in many refinery fields.[10–13]

Partial least-squares regression was employed to build calibration models to evaluate the technical feasibility of on-line monitoring of the LBO process. PLS models were developed using only second-derivative spectra because these can help to enhance the spectral features and remove (or at least reduce) baseline variation, as shown in Fig. 2. Two different spectral ranges, 1100–1580 nm and 1100–1580 nm combined with 1870–2140 nm, were used to build the calibration models. The absorption bands from 1580–1870 and 2140–2500 nm were excluded due to the saturation of NIR radiation, as previously described. Both PLS calibration models were also developed with the use of reference pour-point analysis from the standard 3 °C (designated as Experiment A) and the narrower 1 °C decrease interval (designated as Experiment B), as described in the Introduction.

The optimum number of factors was identified as the number of factors that gives a minimum MSECV (mean square error of cross validation). Cross-validation was used as a validation method by dividing the calibration data set into four segments. The MSECV is a good index to find the optimal number of PLS factors without overmodeling. Figure 3 shows the MSECV plotted as a function of the number of PLS factors used to determine the pour point within the 1100–1580 nm region (Experiment B). As expected, the MSECV decreases very sharply for the first factor and gradually decreases as more pour-point-dependent spectral variation is incorporated into the calibration model. However, from the number of factors of 4, no significant improvement in MSECV is observed. This indicates that the data has been over-fit by incorporating spectral information into the model that is not related to the pour point. In this case, 3 PLS factors were chosen as optimum. The optimal number of PLS factors for the 1100–1580 nm range combined with the 1870–2140 nm range was also determined by generating and examining the same type of plot.

Table I shows the results of the PLS calibration using pour point analysis from the standard 3 °C (Experiment A) and narrower 1 °C decrease interval (Experiment B). As shown in the table, Experiment B provides the better calibration performance since the reference lab data are more accurate, as expected. MSECV is greatly improved with Experiment B compared to Experiment A. Similar calibration results were achieved using both spectral ranges in each experiment. The 1100–1580 nm range contains enough spectral information for pour-point (refer to Fig. 2) determination, so the inclusion of the extra 1870–2140 nm range does not help to improve calibra-

TABLE I. PLS calibration results (using second-derivative spectra) for the measurement of the pour point of lube base oil. Experiment A and B correspond to the reference pour-point measurement at 3 and 1 °C decrease intervals, respectively.

	Experiment A		Experiment B	
Spectral range (nm)	Number of PLS factor	MSECV (°C)	Number of PLS factor	MSECV (°C)
1100–1580	3	1.70	3	1.17
1100–1580 and 1870–2140	3	1.66	3	1.14

tion performance significantly. Hence, this range was dropped from further discussion.

To investigate in greater depth the better calibration performance of Experiment B, the PLS score plots of both Experiment A and B were examined. The main purpose was to elucidate whether the better accuracy of Experiment B was from the more descriptive loading vector from the more precise lab analysis or mainly from lab accuracy only. Figure 4 shows the score plot using the first and second scores of Experiment A and B. Open circles and dark diamonds correspond to Experiment A and B, respectively. As shown, scores from both Experiment A and B are very similar. This means that both loading vectors are almost identical. Regardless of reference lab accuracy, both loading vectors to describe pour-point variation are similar to each other. Consequently, the better PLS calibration performance of Experiment B is mainly from the improvement of the reference lab accuracy. The more precise lab measurement provides the more accurate PLS calibration performance.

The correlation plot for both Experiment A and B is shown in Fig. 5. Open circles and dark diamonds represent Experiment A and B, respectively. As stated, for Experiment B, the data has better correlation with the reference pour-point data. The overall results show that the pour point of LBO can be successfully monitored by NIR spectroscopy. When the more accurate reference analysis is available, it clearly shows that NIR calibration performance can be greatly improved. Fortunately, the on-line pour-point analyzer can measure a sample in a continuous mode (not discrete decrease interval), so a more accurate NIR prediction performance is expected when the calibration model is built using an on-line pour-point analyzer.

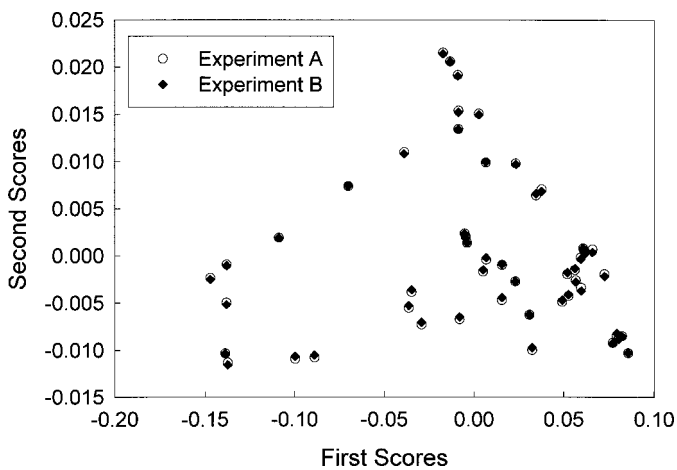

FIG. 4. Score plot using the first and second scores of Experiment A and B.

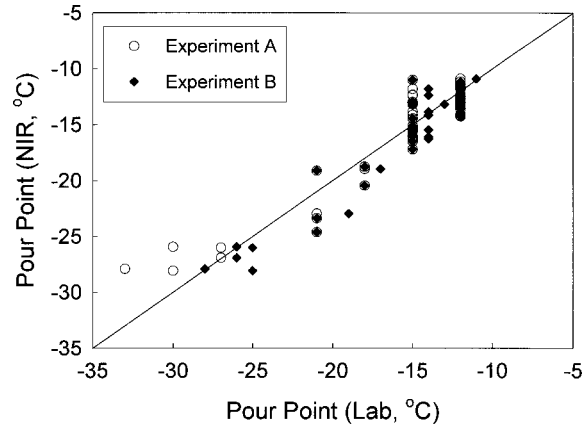

FIG. 5. Pour-point correlation plot for Experiment A and B.

There will be no significant differences between lab and on-line NIR measurements except for the temperature of the sample measurement. Some LBO products are fairly viscous at room temperature, so temperature could influence the flow rate in the sampling system. The temperature of the expected sampling point at the actual LBO process is 50~55 °C; therefore, we decided to measure on-line samples at 50 °C. In this way, it was not necessary to severely control the temperature of the sample from the process temperature, and reasonable flow can be achieved for more viscous samples.

On-Line Near-Infrared Analyzer System. The NIR analyzer system in conjunction with the sampling system was installed for real-time monitoring of the LBO process. The schematic diagram of the overall on-line analyzer system is shown in Fig. 6. As shown, the NIR analyzer was installed in parallel with the conventional existing on-line pour-point analyzer. In this way, NIR analytical performance can be compared with the conventional method in an on-line manner. The fast loop was designed to transport sample to the NIR measurement cell in less than one minute. The overall flow of the fast loop was controlled by the fast-loop regulator. Then, by using the designed sampling conditioning system (SCS), the flow rate of the sample at the NIR measurement cell was regulated at 100 cc/min. Also, the sample temperature was controlled at 50 °C using an electrical heating system. The process NIR spectrometer

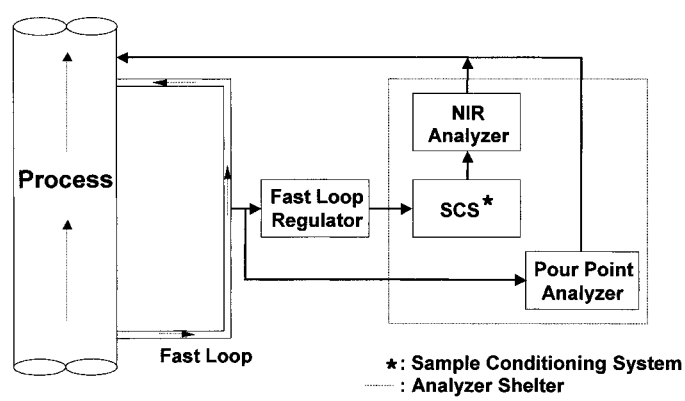

FIG. 6. Schematic diagram of overall on-line analyzer system.

TABLE II. PLS calibration results using on-line NIR spectra and reference data from the on-line pour-point analyzer.

Spectral range (nm)	Number of PLS factor	MSECV (°C)	SEP (°C)
1100–1580	5	0.71	0.67

(NEMA 4 specification) was manufactured by Foss NIR-Systems (on-line model 5000, Silver Spring, MD) and equipped with a quartz halogen lamp, PbS detector, and fiber-optic transflectance probe. The measurement configuration is similar to that described in the Experimental section. The enclosure of the NIR analyzer was also installed to meet the area safety specifications (Class I, Division II). After NIR measurement, the sample returned to the process automatically by pressure difference between sampling and returning points of the process.

After installation, on-line NIR as well as pour-point analyzers were continuously operated to set up the calibration data set. The spectral collection scheme of the process on-line NIR analyzer is slightly different from that of the lab NIR instrument. Lab NIR uses air as the background, while the process analyzer uses separate optical fibers as a background. Therefore, background spectra corresponding to separate reference optical fibers were automatically collected before the collection of every sample single-beam spectra. The NIR spectra, when pour points were determined, were selected for the calibration data set. Overall, 47 spectra for the on-line calibration data set were established over a 45-day period. Only one sample spectrum was selected during the production of a specific grade, since there is no significant chemical variation in this period. Even with more samples during the production, there will be no significant compositional variations. Usually, the production switch of different grades of LBO product was accomplished every day, so generally one NIR spectrum was selected per day during individual LBO production. Samples were collected over the long period of time to make the data set compositionally wide enough.

Table II shows the PLS calibration results using on-line NIR spectra (second-derivative spectra) and reference data from the on-line pour-point analyzer, which is more precise compared to the lab analyzer. The same cross-validation (4 segments) was used as an internal validation. In this case, only 1100–1580 nm was utilized since we know from the lab feasibility study that this range provided similar accuracy in comparison to that from the 1100–1580 nm range combined with the 1870–2140 nm range. Additionally, absorbance in the 1870–2140 nm range is fairly high (around 1.0 absorbance unit), so this could result in nonlinearity (unstable prediction) when the baseline is slightly shifted towards higher absorption. The resulting MSECV is greatly improved since the on-line pour-point analyzer provides continuous pour-point measurement, which is more precise. The correlation plot of the resulting calibration is shown in Fig. 7 (open circles). Open circles correspond to the calibration data. The resulting MSECV is 0.71 °C when calibrated with the on-line pour-point analyzer.

The on-line prediction performance has been tested over 45 days using the developed model described in Table II. As for collecting calibration NIR spectra, the

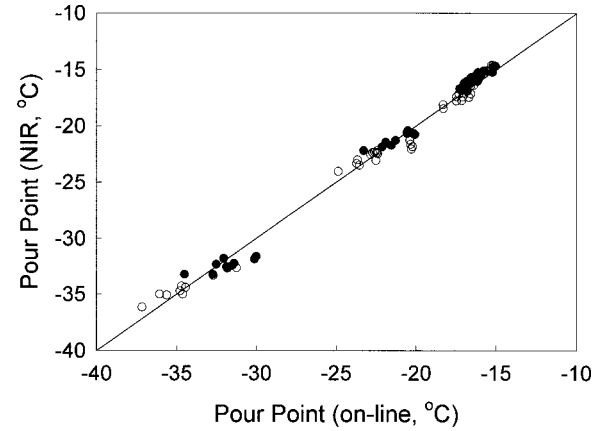

FIG. 7. Pour-point correlation plot for on-line calibration (open circles) and prediction data (filled circles).

NIR prediction results were directly compared with those of the pour-point analyzer. A total of 45 samples were evaluated. The prediction results are shown in Fig. 7 (filled circles) and Table II. An excellent SEP (standard error of prediction) is achieved since the NIR analyzer was calibrated with the more precise and accurate on-line pour-point analyzer. This result clearly shows that lab accuracy is highly important for the performance of a given spectroscopic system, especially when chemometric calibration is involved.

CONCLUSION

Using NIR spectroscopy, the pour point of LBO has been successfully monitored in an on-line manner. In comparison with the conventional on-line pour-point analyzer, NIR spectroscopic analysis provides much faster measurement with similar precision. This real-time analysis enables the monitoring of process variation much more quickly and with clearer results. Then necessary process control and optimization technology can be applied to achieve real economical benefits such as increased throughput and maximization of process efficiency. Future work will be directed to develop NIR calibration models for other LBO properties, such as viscosity, viscosity index, and Noack volatility. For this purpose, thorough understanding of the chemistry related to these physical properties will be necessary.

ACKNOWLEDGMENT

This work is supported by the Korea Science and Engineering Foundation (Grant Number: R14-2002-004-01000-0).

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